

GENERAL SPECIFICATION G1.6

Materials, Plant and Equipment

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G1.6 MATERIALS, PLANT AND EQUIPMENT

G1.6.1 Materials - General Requirements

- (a) The term materials shall mean all materials, goods and article, of every kind whether raw, processed or manufactured and equipment and plant of every kind, to be supplied by the Contractor for incorporation into the Works.
- (b) Except as may be otherwise specified for particular parts of the Works where clarified in in the Particular Specification, the provisions of General Specification shall apply to materials general requirements for any part of the Works.
- (c) All materials shall be new and of the types and qualities described in the Contract and shall be at least equal to approved samples.
- (d) The Contractor shall as far as possible utilize materials available in Singapore in the execution and completion of the Works, subject to compliance with the Specification.
- (e) Materials shall be transported, handled and stored in such a manner as to prevent deterioration, damage or contamination.

G1.6.2 Substances and Products

- (a) Substances and products used in the Works which may be applied to or introduced into water which is to be supplied for drinking, washing or cooking shall not contain any matter which could impart taste, odour, colour or toxicity to the water or otherwise be objectionable on health grounds. This applies to potable water treatment and NEWater treatment and conveyance applications.
 - (i) Only those substances and products which meet the requirements of paragraph (2) of Regulation 31 of the Water Supply (Water Quality) Regulations 2000, UK, or have the approval of one of the following international regulatory bodies shall be used in such situations:
 - 1) American National Standards Institute Standard 61, Drinking water system components - health effects;
 - 2) National Sanitation Foundation (USA) Standard 60, Drinking water chemicals - health effects; and
 - 3) Regulatory bodies of France, Germany or Netherlands.
 - (ii) In the instances where membrane bioreactor based pre-treatment is used upstream of RO system for NEWater production, compliance to the above regulations shall be applied from the membrane filtrate tanks, including all other downstream processes involved in NEWater production and conveyance.
- (b) After award of Contract, from time to time the Contractor shall submit for the approval of the S.O. a schedule of substances and products he proposes to use in the Works giving the following information as applicable: -
 - (i) Item of plant;
 - (ii) Substance/product in contact with water;
 - (iii) Manufacturer of plant/substance/product;

- (iv) Point of use in the Works;
 - (v) Name of the regulatory body, which has approved the substance/product;
 - (vi) Date of approval (for use on water quality and on health grounds); and
 - (vii) Approval (with respect to water quality and health) reference number.
- (c) For all other products and substances to be used in the Works in such situations, the Contractor shall at his own expense obtain the necessary approval and follow the procedure set above and obtain the approval of the S.O. before use.

G1.6.3 Plant, Equipment and Materials

- (a) All manufacturers for Plant, equipment and materials must operate a quality system compliant with ISO 9001:2000 (or equivalent) as minimum to assure the conformance to the highest quality standards of the industry.
- (b) All Plant, equipment and materials furnished by the Contractor shall be made available for inspection and approval of the S.O.. No materials shall be delivered to the Site without the prior approval of the S.O. through the use of the submittal procedures.
- (c) Where materials or articles are specified to comply with standards, the Contractor shall provide the appropriate certificate of compliance, if required by the S.O..
- (d) The Contractor shall submit to the S.O. data (using the BIM model where feasible) relating to the Plant, equipment and materials he proposed to furnish for the Works. Such data shall be sufficient detail to enable the S.O. to identify the particular product and to form an opinion as to its conformity with the Specifications.
- (e) Facilities and labour for the handling and inspection of all Plant, equipment and materials shall be furnished by the Contractors. If the S.O. requires, either prior to beginning or during the progress of the Works, the Contractor shall submit samples of materials for inspections, such tests as may be necessary to demonstrate that they conform to the Specifications. Such samples shall be furnished, stored, packed and shipped as directed at the Contractor's expenses.
- (f) The Contractor shall submit data and samples of material sufficiently early to permit confirmation and approval before materials are needed for incorporation in the Works; the quality of the samples representing the quality of the bulk of such materials which he is proposing for use. Any delay of approval resulting from the Contractor's failure to submit samples or data promptly shall not be used as a basis of claim against the Board.
- (g) In order to demonstrate the proficiency of workmen, the Contractor shall provide samples of workmanship when requested by the S.O..
- (h) All Plant, equipment and materials supplied and installed shall not contain any polychlorinated biphenyls (PCB), particularly in capacitors for motors and lamps.
- (i) The Contractor shall ensure that the Plant is safe in use, and that any safety regulations imposed by law or by any authorized body empowered to make such regulations are complied with.

G1.6.4 Noise

- (a) No item of Plant shall produce a sound pressure level exceeding 85 dB (A) when measured at a distance of 1 m from the reference surface of that item in a horizontal direction to comply with Singapore WSH (noise) Regulations 2011.
- (b) For the purposes of sound pressure level measurement, any item of plant shall include the driver, driven equipment, pipework, ductwork, any associated equipment and all other attachments that may produce noise.
- (c) The Contractor shall be responsible for carrying out all sound pressure level measurements on Site.
- (d) Sound pressure levels shall be measured in dB(A), using a calibrated sound meter meeting the requirements of BS EN 61672-1 and BS EN 61672-2.
- (e) For major items of Plant, the Contractor shall provide certificates from the manufacturer covering sound pressure level tests carried out in the factory or type test certificates for a similar item of Plant.
- (f) If an item of Plant/equipment in its standard build does not comply with the above requirement, the manufacturer shall reduce the sound pressure level by providing improved or additional silencers or fitting sound insulating materials to the item until the above requirement has been complied with, at no additional cost. In general, acoustic hoods or similar will not be accepted.
- (g) The S.O. intends to require another contractor to construct a noise model of TWRP. The Contractor shall provide the measured sound pressure levels for all Plant and equipment for input to the noise model.

END OF SECTION